Customer Shopping Behavior Analysis

End-to-end analysis of customer purchase activity using

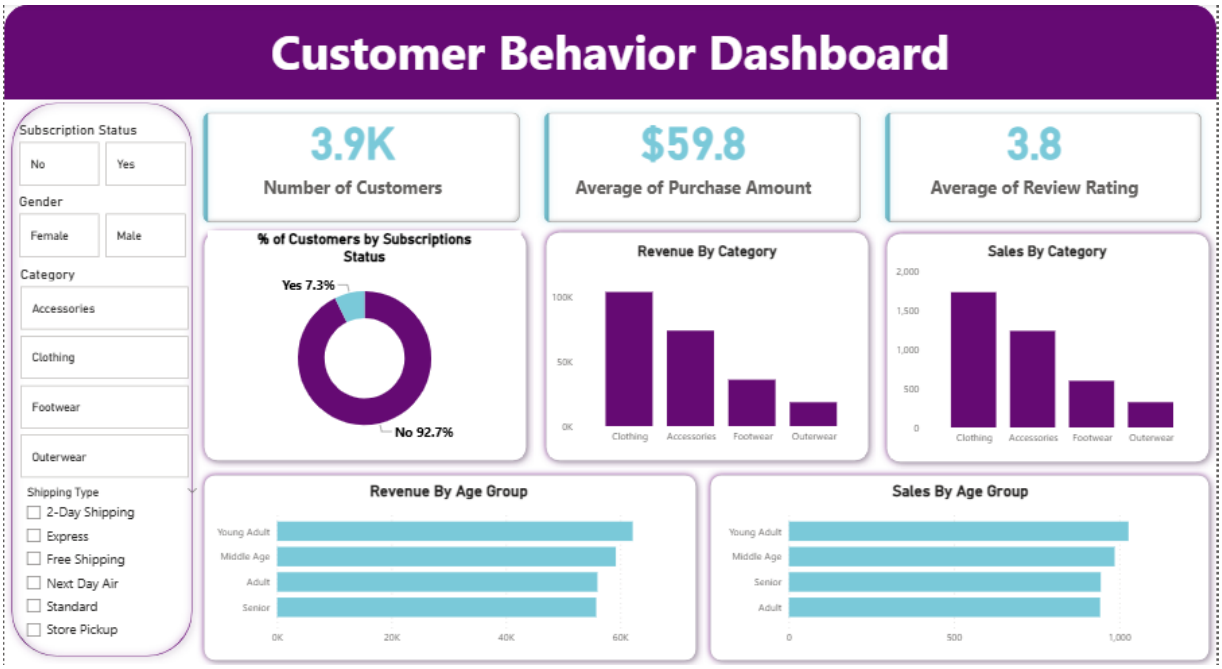
Python, PostgreSQL and Power BI.

1. Executive Summary:

This project examines 3,900 customer transactions across four retail categories to understand how demographics, purchase patterns, subscriptions and product choices influence revenue.

The workflow combines Python for cleaning and feature engineering, PostgreSQL for business-oriented analysis, and Power BI for dashboarding and interactive exploration.

Key findings point to strong revenue concentration in Clothing, very low subscription adoption, and a high-value Young Adult segment that should be prioritized in retention and acquisition campaigns.



2. Dataset Snapshot:

Attribute	Detail
Total records	3,900 rows
Total features	18 columns
Missing data	37 null values in review_rating
Categories	Clothing, Accessories, Footwear, Outerwear
Age range	18 to 70 years

Purchase range	$20 to $100

Feature families: Demographics (age, gender, location, subscription status), purchase details (item, category, amount, season, size, color) and shopping behavior (discount usage, prior purchases, review rating, shipping type).

3. Data Preparation in Python:

3.1 Data cleaning

- The dataset was profiled with pandas using structure and descriptive summaries to identify nulls, column types and value distributions.
- The 37 missing review ratings were imputed with a per-category median instead of a global median, preserving rating behavior within each product group.

3.2 Standardization

- Columns were renamed to snake_case to ensure consistent naming across Python, SQL and Power BI.
- The promo_code_used field was removed after checking that it duplicated the signal already captured by discount_applied.

3.3 Feature engineering

- A new age_group feature was created with quantile-based bins to support segment comparisons across balanced age bands.
- purchase_frequency_days was derived from the original frequency labels to enable numeric analysis of repeat buying behavior.

3.4 Database loading

- The cleaned dataframe was pushed into PostgreSQL through SQLAlchemy to support reusable SQL queries and dashboard-ready summary tables.

Age group	Range	Count
Young Adult	18–31	1,028
Adult	32–44	942
Middle Age	45–57	986
Senior	58–70	944

4. SQL Findings and Business Insights

Ten SQL questions were used to evaluate revenue concentration, customer segments, product performance and subscription behavior. The most relevant findings are summarized below.

Revenue by gender: Male customers generated $157,890 versus $75,191 from female customers, largely because the male segment contains more customers rather than a radically higher ticket size.	**Subscribers vs non-subscribers:** Average spend per transaction is nearly identical ($59.49 vs $59.87), but only 1,053 customers are subscribed compared with 2,847 non-subscribers.

Selected result highlights

- 839 customers used a discount and still spent above the global average purchase amount, indicating a promotion-responsive but valuable segment.

• Top-rated products were Gloves, Sandals, Boots, Hat and Skirt, with ratings concentrated in a narrow 3.78 to 3.86 band.

• Young Adults generated the highest revenue ($62,143), ahead of Middle Age ($59,197), Adult ($55,978) and Senior ($55,763).

• Loyal customers dominate the base (3,116), while only 83 are classified as New, suggesting strong retention but weak acquisition.

5. Recommendations:

	Recommendation	Why it matters
1	**Strengthen the subscription offer**	Only 7.3% of customers are subscribed; better perks and targeting could convert repeat buyers.
2	**Build a loyalty program around the loyal segment**	With 3,116 loyal customers, retention actions could compound value quickly.
3	**Audit heavy discount products**	Some items approach a 50% discount rate, which may pressure margin without guaranteed lift.
4	**Prioritize Young Adults in campaigns**	This segment contributes the highest revenue while group sizes remain relatively balanced.
5	**Feature top-rated products in promotions**	High-rated items such as Gloves, Sandals and Boots are strong candidates for hero placements.
6	**Use express-shipping buyers for upsell tests**	Express customers spend slightly more on average, suggesting stronger buying intent.

6. Tools Used:

Tool	Purpose
Python / pandas	Cleaning, preprocessing and feature engineering
PostgreSQL	Business analysis with SQL queries
SQLAlchemy	Data loading from Python into PostgreSQL
Power BI	Interactive dashboard and storytelling visuals
Kaggle dataset	Source of the customer shopping behavior records